VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Belagavi: 590 018

Report on

"Activity on Creating Effective Prompts

for

Introduction to AI and its Applications (1BAIA103)

Submitted By

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DATE	MAX MARKS	MARKS SCORED	FACULTY SIGN

Activity 1: Basic Prompt writing: Create two different prompts to ask an AI about the topic "Electricity." The first prompt should be vague, and the second prompt should be clear and specific. Compare the responses you get and describe which prompt gave a better answer and why.

Vague Prompt:

Prompt:

"Tell me about electricity."

Sample AI Response (Vague Prompt):

Electricity is a form of energy that comes from the movement of electrons. It is used to power homes, machines, and electronic devices. Electricity can be produced from different sources like coal, water, wind, and solar energy. It is very important in daily life.

Clear & Specific Prompt:

Prompt:

Explain electricity for a high school student, including its definition, basic types (AC and DC), and one daily-life example for each."

AI Response (summary):

Electricity is the flow of electric charges. There are two types: AC (used in homes for fans and lights) and DC (used in batteries like mobile phones).

Comparision

Conclusion

2. Zero-Shot Prompting: Create a prompt that asks an AI to explain Ohm's Law

without giving any example or background. Evaluate how well the AI explains the concept based on your prompt alone.

Zero-Shot Prompt:

Prompt:

"Define Ohm's Law and explain the relationship between voltage, current, and resistance using proper scientific terms."

AI Response (Expected)

Ohm's Law states that the current flowing through a conductor is directly proportional to the potential difference across it, provided the temperature and physical conditions remain constant. Mathematically, it is expressed as **V = IR**, where **V** is voltage, **I** is current, and **R** is resistance. The law explains how resistance controls the flow of electric current in an electrical circuit.

Evaluation

The AI explains Ohm's Law clearly and accurately without requiring any examples or prior background. The use of correct scientific terminology and the standard formula shows strong conceptual understanding. However, since no examples or applications are included, the explanation remains theoretical. Despite this limitation, the AI successfully conveys the core concept based solely on the zero-shot prompt, demonstrating effective zero-shot learning capability.

Comparision

Conclusion

3. One-Shot and Few-Shot Prompting: Provide the AI with a single example of how to calculate the resistance in a simple circuit. Then write your own prompt asking the AI to solve a similar resistance calculation. After that, add two more examples to your prompt and observe any changes in the AI's response quality.

One-Shot Prompting:

Single Example Given to the AI

Example:

"If the voltage across a circuit is 48 V and the current is 8 A, the resistance is calculated using Ohm's Law:

R = V / I = 48 / 8 = 6 Ω."

My Prompt to the AI (One-Shot)

"Using the example above, calculate the resistance when the voltage is40 V and the current is 4 A."

AI Response (Observed)

The AI correctly applies Ohm's Law:

R = V / I = 40 /4=10 Ω.

The response is accurate but short, closely following the given example.

Few-Shot Prompting

Additional Examples Added

Example 1:

V = 10 V, I = 2 A

R = 10 / 2 = 5 Ω

Example 2:

V = 9 V, I = 3 A

R = 9 / 3 = 3 Ω

Updated Prompt (Few-Shot)

"Using the pattern shown in the above examples, calculate the resistance when the voltage is 24 V and the current is 6 A. Show the formula and final answer."

AI Response (Observed)

The AI clearly states the formula, substitutes values, and gives the final result:

$R = V / I = 24 / 6 = 4\ \Omega$.

The explanation is more structured and step-by-step.

Comparision

Conclusion

4. Chain-of-Thought Prompting: Develop a prompt that guides the AI step-by-step through calculating current flow in a circuit using Ohm's Law with resistors in series. Then, ask a final question for the AI to solve. Analyze how breaking down the reasoning steps impacts the accuracy of the answer.

Chain-of-Thought Prompting: Current Calculation in Series Circuit

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Step-by-Step Prompt Given to the AI

"Follow these steps to calculate the current in a series circuit:

1. Identify the given voltage and resistances.
2. Add all resistor values to find the total resistance (since they are in series).
3. Apply Ohm's Law $I = \frac{V}{R}$
4. Clearly show each step and the final answer.

Now solve this problem:
A circuit has a voltage supply of 18 V and three resistors in series with values 4 Ω, 8Ω, and 2Ω. Calculate the current flowing through the circuit."

AI Answer (Observed)

- Total resistance = 4 + 8 + 2 = 14 Ω
- Using Ohm's Law:
  $I = \frac{14}{14} = 1\text{A}$
- Current flowing in the circuit is **1 A**.

Analysis

Breaking the problem into clear reasoning steps helps the AI follow a logical path instead of jumping directly to the answer. This reduces the chance of calculation or conceptual errors, especially in multi-step problems like series circuits. The structured prompt improves accuracy, clarity, and consistency of the response.

Comparision

Conclusion

5. Prompt Refinement: Start with an ambiguous prompt related to the "Water Cycle." Test the AI's response, note the confusion or errors, and then refine your prompt to make it clearer and more specific. Repeat this process twice and record how the AI's responses improve with each refinement.

Role-Based Prompting: Create three prompts asking the AI to explain "Newton's Laws of Motion," each with a different role instruction: (a) as an expert engineer, (b) as a high school teacher, (c) as a beginner. Compare the tone, detail, and style of the responses.

Part 1: Prompt Refinement (Water Cycle)

Initial Ambiguous Prompt

"Explain the water cycle."

AI Response (Observed):

The AI gives a very general explanation mentioning evaporation, condensation, and precipitation, but it lacks structure, sequence, and depth. Some processes like transpiration or collection are briefly mentioned or missing.

Issue Noted:

The prompt is too broad, so the response is basic and not well-organized.

Refinement 1

"Explain the stages of the water cycle in order."

AI Response (Observed):

The AI now explains evaporation, condensation, precipitation, and collection in a logical sequence. The explanation is clearer but still brief and not detailed.

Improvement:

Better structure and correct ordering of processes.

Refinement 2 (Clear & Specific Prompt)

"Explain the water cycle for a middle school student, describing evaporation, condensation, precipitation, transpiration, and collection with simple definitions."

AI Response (Observed):

The AI provides a well-structured, easy-to-understand explanation covering all major stages with clear definitions and smooth flow.

Final Improvement:

The response becomes more complete, accurate, and audience-appropriate.

Conclusion (Prompt Refinement)

As the prompt becomes clearer and more specific, the AI's response improves in structure, accuracy, and usefulness. Refining prompts helps reduce ambiguity and produces higher-quality answers.

Part 2: Role-Based Prompting (Newton's Laws of Motion)

(a) Role: Expert Engineer

Prompt:

"Explain Newton's Laws of Motion as an expert mechanical engineer."

Response Style (Observed):

The explanation is technical, precise, and application-focused. It includes real-world engineering examples and emphasizes forces, mass, acceleration, and system behavior.

(b) Role: High School Teacher

Prompt:

"Explain Newton's Laws of Motion as a high school physics teacher."

Response Style (Observed):

The explanation is clear, structured, and student-friendly. Simple language and classroom examples are used, focusing on understanding rather than complexity.

(c) Role: Beginner

Prompt:

"Explain Newton's Laws of Motion to a complete beginner."

Response Style (Observed):

The explanation is very simple, conversational, and intuitive. Complex terms are avoided, and the focus is on basic ideas.

Comparision

Conclusion

6. Creative Engineering Problem Prompts: Craft a prompt that asks the AI to brainstorm ideas for designing a low-cost water purification system suitable for rural areas. Encourage creativity by adding phrases like "limited resources" and "sustainability".

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Comparision

Conclusion

7. Ethical Prompt Design Discussion: Identify a biased prompt related to job descriptions (e.g. language with respect to a gender). Rewrite the prompt to remove bias and create a neutral, inclusive version. Explain why this revision is more ethical.

Biased Prompt (Example)

"Write a job description for a **young, aggressive salesman** who can dominate clients and work long hours without family distractions."

Issues with the Prompt

- Uses **gender-coded and exclusionary language** ("salesman").
- Implies **age bias** ("young").
- Encourages a **hostile personality stereotype** ("aggressive").
- Assumes personal circumstances ("without family distractions"), which is discriminatory.

Rewritten Neutral & Inclusive Prompt

"Write a job description for a **sales professional** who can build strong client relationships, meet performance goals, and work effectively in a team-oriented environment."

Why the Revised Prompt Is More Ethical

- Uses **gender-neutral language** ("sales professional").
- Removes **age and personal-life assumptions**.
- Focuses on **skills, responsibilities, and performance**, not stereotypes.
- Promotes **fair access and inclusivity** for all qualified candidates.

Comparision

Conclusion

8. Simulated Customer Support Chatbot: Develop a prompt that instructs the AI to play the role of a technical support agent helping a customer troubleshoot a failure in an electronic circuit. Include instructions to keep the tone friendly and professional and to ask diagnostic questions.

Simulated Customer Support Chatbot – Prompt Design

Prompt:

"You are a technical support agent for an electronics company. Your role is to help a customer troubleshoot a non-working electronic circuit. Communicate in a friendly, calm, and professional tone. Ask clear diagnostic questions step by step to understand the problem before suggesting solutions. Guide the customer using simple explanations, avoid technical jargon unless necessary, and ensure the customer feels supported throughout the conversation."

Why This Prompt Works

- **Role clarity:** Clearly defines the AI as a technical support agent.
- **Tone guidance:** Ensures friendly and professional communication.
- **Problem-solving approach:** Encourages asking diagnostic questions before jumping to conclusions.
- **User-focused:** Emphasizes clarity, patience, and customer satisfaction.

Outcome

This prompt helps the AI simulate a realistic customer support chatbot that can effectively diagnose electronic circuit issues while maintaining a positive and helpful interaction style

Comparision

Conclusion

9. Multi-Language Prompting: Develop a prompt that asks the AI to translate a simple engineering glossary (5 technical terms) from English to your native language. Then modify the prompt to request additional explanations of these terms in the translated language.

Prompt 1: Translation Only

"Translate the following engineering terms from **English to kannada**

1. **Force** – ಶಕ್ತಿ (Shakti)
2. **Energy** – ಶಕ್ತಿ ಅಥವಾ ಉರ್ಜಾ (Shakti / Urja)
3. **Work** – ಕೆಲಸ (Kelasa)
4. **Pressure** – ಒತ್ತಡ (Ottada)
5. **Efficiency** – ಕಾರ್ಯಕ್ಷಮತೆ (Karyakshamate

AI Response (Expected – Summary)

The AI provides accurate Hindi equivalents for each engineering term using standard scientific vocabulary.

Prompt 2: Translation with Explanation

"Translate the following engineering terms from **English to kannada** and also explain each term in **simple kannada**, using short and clear definitions suitable for beginners

1. **Force (ಶಕ್ತಿ – Shakti)**
   - ಶಕ್ತಿ ಎಂದರೆ ಯಾವುದಾದರೂ ವಸ್ತುವನ್ನು ತಳ್ಳುವುದು ಅಥವಾ ಎಳೆಯುವುದು.
   - ಉದಾಹರಣೆ: ಕಪ್ಪು ಬಲವಾಗಿ ಒತ್ತಿದಾಗ ಅದು ಮುಂದೆ ಸಾಗುತ್ತದೆ.
2. **Energy (ಉರ್ಜಾ – Urja)**
   - ಉರ್ಜಾ ಎಂದರೆ ಕಾರ್ಯ ಮಾಡಲು ಬೇಕಾದ ಸಾಮರ್ಥ್ಯ.
   - ಉದಾಹರಣೆ: ಬ್ಯಾಟರಿ ಉರ್ಜಾ ಹೊಂದಿರುವುದರಿಂದ ಲೈಟ್ ಬೆಳಗುತ್ತದೆ.
3. **Work (ಕೆಲಸ – Kelasa)**
   - ಕೆಲಸ ಎಂದರೆ ಶಕ್ತಿ ಬಳಸಿ ವಸ್ತುವನ್ನು ಚಲಿಸುವಂತೆ ಮಾಡುವ ಪ್ರಕ್ರಿಯೆ.
   - ಉದಾಹರಣೆ: ಪುಸ್ತಕವನ್ನು ಮೇಜಿನ ಮೇಲೆ ಹಾಕುವುದು ಒಂದು ಕೆಲಸ.
4. **Pressure (ಒತ್ತಡ – Ottada)**
   - ಒತ್ತಡ ಎಂದರೆ ಏಕಕಾಲದಲ್ಲಿ ಒಂದು ಪ್ರದೇಶಕ್ಕೆ ಬಲ ಹಾಯಿಸುವ ಪ್ರಮಾಣ.
   - ಉದಾಹರಣೆ: ನೀರಿನ ಟ್ಯಾಂಕನಲ್ಲಿ ಹೀಳಿಕೆರ್ ಮೇಲೆ ಒತ್ತಡ ಉಂಟುಮಾಡುತ್ತದೆ.
5. **Efficiency (ಕಾರ್ಯಕ್ಷಮತೆ – Karyakshamate)**

- ಕಾರ್ಯಕ್ಷಮತೆ ಎಂದರೆ ಸರಿಯಾದ ರೀತ್ತಲ್ಲಿ ಶಕ್ತಿ ಬಳಸ್ತವ ಸಾಮರ್ಥ್ಯ.
- ಉದಾಹರಣೆ: ಒಂದು ಲೈಟ್ ಬಲ್ಬ್ ಕಡಿಮೆ ವಿದ್ಯುತ್ ಬಳಸಿ ಹೆಚ್ಚು ಬೆಳಕು ಕೊಡುತ್ತಿದಾದರೆ ಅದು ಕಾರ್ಯಕ್ಷಮವಾಗಿದೆ.

Comparision

Conclusion

10. Review a curated set of different prompt types (e.g., for summarization, information extraction, paraphrasing, question answering) from a "Prompt Gallery." For each prompt type, match it with a real-world task (e.g., summarizing a lecture note, extracting names from a project report). Test at least three prompt templates on an AI tool or by role-play (students simulate being the AI), with varied wording. Record the outcomes and discuss which prompt (or template) was most effective for each task, and explain why you think it worked best. Reflect on how changing small parts of a prompt can alter model response quality, completeness, or accuracy.

Prompt Gallery Review and Testing

1. Summarization

Prompt Template:

"Summarize the following text in 3–5 sentences, keeping only the key points and main ideas."

Real-World Task: Summarizing lecture notes for quick revision.

Testing/Outcome:

- Original Lecture Note Excerpt: "Photosynthesis is the process by which green plants and some other organisms use sunlight to synthesize foods from carbon dioxide and water. It generally involves the green pigment chlorophyll and generates oxygen as a by-product."
- AI Output (using template): Summarizes in 3 sentences: main idea about green plants using sunlight to make food, role of chlorophyll, and oxygen as by-product.
- Effectiveness: High – concise, accurate, includes all essential points.

Observation: Changing "3–5 sentences" to "in one sentence" reduced completeness, while asking "in your own words" improved readability and reduced copy-paste phrasing.

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2. Information Extraction

Prompt Template:

"Extract all names of people, organizations, and locations mentioned in the text and list them separately."

Real-World Task: Extracting contributors and institutions from a project report.

Testing/Outcome:

- Project Report Excerpt: "The AI project was conducted by Aloke Reddy at Tech Innovations Pvt. Ltd., with guidance from Dr. Meera Sharma at Bangalore University."
- AI Output:
    - People: Aloke Reddy, Dr. Meera Sharma
    - Organizations: Tech Innovations Pvt. Ltd., Bangalore University
    - Locations: Bangalore
- Effectiveness: Very high – clearly categorized and correct.

Observation: Wording changes like "List only names without context" produced slightly less informative outputs; adding "include designation/title if mentioned" improved clarity and usefulness.

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3. Paraphrasing

Prompt Template:

"Rewrite the following paragraph in simpler language without changing its meaning."

Real-World Task: Simplifying technical content for beginner students.

Testing/Outcome:

- Original Text: "The mitochondria is often called the powerhouse of the cell because it produces energy in the form of ATP through respiration."
- AI Output: "Mitochondria are called the cell's powerhouse because they make energy (ATP) through respiration."
- Effectiveness: High – meaning preserved, simpler wording, easier to understand.

Observation: Adding "use casual language" made the text even more relatable, while "make it formal" maintained technical tone but slightly harder for beginners.

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4. Question Answering

Prompt Template:

"Answer the following question based on the text. Provide a clear and concise response."

Real-World Task: Answering comprehension questions from a textbook excerpt.

Testing/Outcome:

- Text Excerpt: "Newton's First Law states that an object will remain at rest or in uniform motion

unless acted upon by an external force."

- Question: "What does Newton's First Law explain?"
- AI Output: "It explains that an object stays at rest or keeps moving uniformly unless an external force acts on it."
- Effectiveness: High – accurate, concise, and easy to understand.

Observation: Adding "explain with an example" produced longer, more illustrative answers. Removing "concise" sometimes led to unnecessarily lengthy responses**.**

Reflection on Prompt Variations

1. Small wording changes matter:
    - Adding "in your own words" improves originality.
    - Adding "concise" or "simple" affects length and readability.
2. Context matters:
    - Specifying the audience (students vs experts) changes tone and depth.
3. Templates that guide structure work best:
    - Summarization prompts with clear sentence limits produced the most complete outputs.
    - Information extraction prompts with explicit categories gave accurate results consistently**.**

Comparision

Conclusion

11. Choose a real engineering challenge or societal problem relevant to your field (e.g., "Reducing plastic waste in campus cafeterias" or "Optimizing solar panel placement on campus rooftops"). Draft an initial prompt that asks an AI to propose practical solutions. Share the AI's (or peer's) answer in small groups and identify aspects that are missing, vague, or not actionable. Refine your prompt based on feedback (e.g., specify constraints, ask for step-by-step solutions, or require a list of pros and cons). Repeat the process one more time, refining again for further clarity or specificity. Document the entire prompt-refinement process and share the best solution generated, along with a brief analysis of how prompt improvements led to better responses.

Engineering Challenge:

Optimizing solar panel placement on campus rooftops to maximize energy efficiency.

Step 1: Initial Prompt

Prompt 1 (Initial / Broad):

"Suggest practical ways to place solar panels on campus rooftops to produce more energy."

AI Response (Observed):

- "Install panels on all rooftops."
- "Use high-efficiency panels."
- "Ensure they face sunlight."

Analysis / Issues:

- **Vague:** No details on placement, angles, or energy calculations.
- **Not actionable:** No step-by-step plan, cost considerations, or constraints.
- **Missing info:** Roof orientation, shading, weather impact, or pros/cons of options.

Step 2: First Refinement

Prompt 2 (More Specific):

"Propose practical solutions for placing solar panels on campus rooftops to maximize energy production.

Include:

- Step-by-step placement recommendations
- Consideration of roof orientation, shading, and panel angle
- Estimated advantages and disadvantages of each option"

AI Response (Observed):

1. **Flat Roofs:** Use adjustable mounts to set panels at optimal angles.
    - **Pros:** Maximizes sunlight exposure, flexible angle adjustment.
    - **Cons:** Higher installation cost.
2. **Sloped Roofs:** Place panels along the slope facing south (northern hemisphere).
    - **Pros:** Easy installation, natural sunlight exposure.
    - **Cons:** Limited adjustability.
3. **Shaded Areas:** Avoid placing panels where buildings or trees cause shadows.
4. **Rooftop Spacing:** Maintain gaps between panels to reduce overheating.

Analysis:

- More **actionable and structured**.
- Pros and cons included.
- Still missing some **quantitative guidance**, e.g., number of panels per roof, expected energy output.

Step 3: Second Refinement

Prompt 3 (Highly Refined):

"Design a detailed plan for installing solar panels on campus rooftops to maximize energy efficiency.

Include:

- Step-by-step placement strategy for flat and sloped roofs
- Suggested tilt angles for each roof type based on latitude
- Consideration of shading, panel spacing, and maintenance access
- Estimated energy output or efficiency gain (qualitative if exact numbers unavailable)
- List of pros and cons for each solution
- Cost-effective recommendations where possible"

AI Response (Best Solution – Summary):

1. **Flat Roofs:**

- Install panels on adjustable racks with tilt ~25–30° (based on latitude).
- Keep at least 1-meter spacing between rows for airflow and maintenance.
- **Pros:** Optimized sunlight capture, easy maintenance.
- **Cons:** Higher initial cost.

2. **Sloped Roofs:**
   - Align panels with roof slope facing south; tilt angle = slope angle.
   - **Pros:** Easy to install, low maintenance.
   - **Cons:** Angle cannot be adjusted, slightly less energy optimization.
3. **Shading Considerations:**
   - Avoid shadows from trees, AC units, or neighboring buildings.
   - Consider microinverters for partially shaded panels to improve efficiency.
4. **Energy Output:**
   - Panels on optimized rooftops expected to generate ~20–30% more energy than non-optimized placement.
5. **Cost-Effective Recommendations:**
   - Use standard efficiency panels for less critical rooftops.
   - Reserve high-efficiency panels for high-exposure roofs to maximize return on investment.

Analysis of Prompt-Refinement Process

- **Initial Prompt:** Too broad → AI gave vague suggestions.
- **First Refinement:** Added step-by-step guidance and pros/cons → response became structured and actionable.
- **Second Refinement:** Added angles, shading, energy output, and cost guidance → response became highly practical, detailed, and suitable for implementation.

Key Insight: Each refinement narrowed scope, added measurable or actionable constraints, and clarified expectations. Small changes like specifying angles, spacing, or cost considerations significantly improved the **quality, usability, and completeness** of the AI's recommendations.

Comparision

Conclusion